

From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

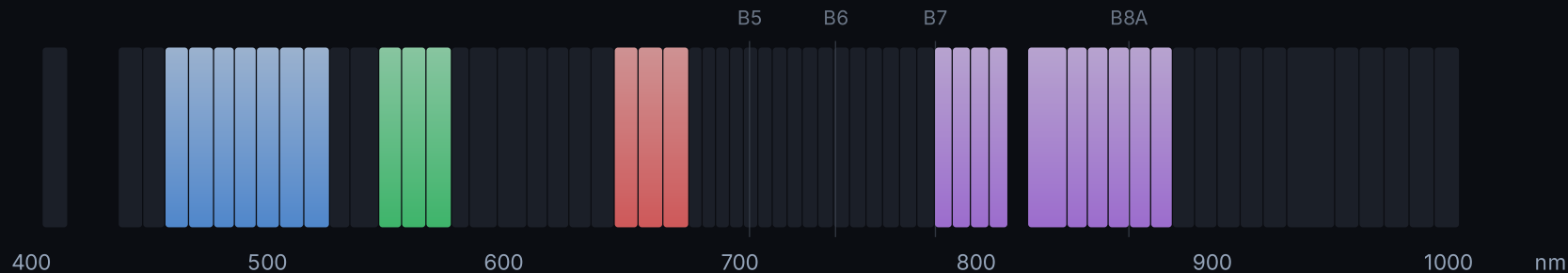
which each CHRIS band goes to the Sentinel-2 band it overlaps the most

how plain **mean** of the CHRIS bands assigned to each pseudo-band

then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

Training mapping, the one the model actually learned. The HuggingFace loader will be aligned to this.

Mode 1 : 62 bands, ~34 m



7 bands, 106%

B2

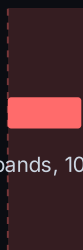
459-525 nm



3 bands, 86%

B3

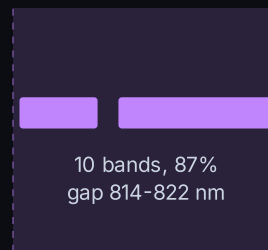
542-578 nm



3 bands, 102%

B4

649-680 nm



10 bands, 87%
gap 814-822 nm

B8

780-886 nm

Mode 2 : 18 bands, ~17 m



B2
459-525 nm



B3
542-578 nm

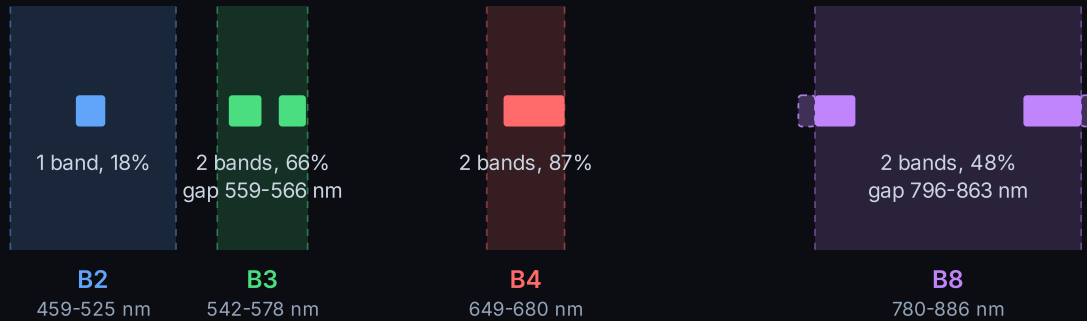


B4
649-680 nm

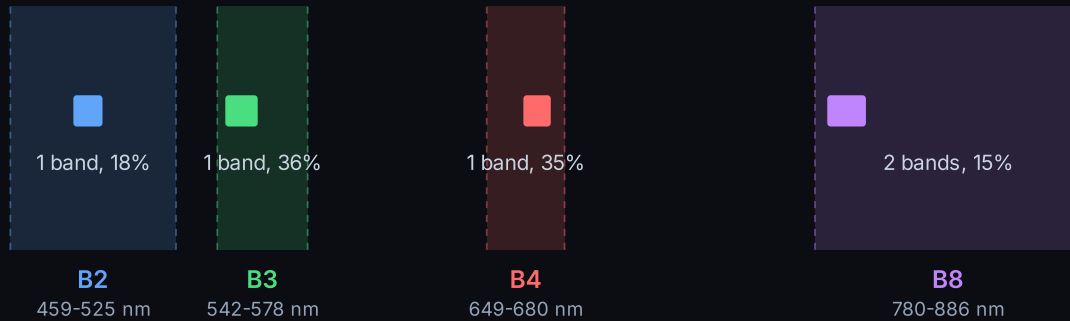
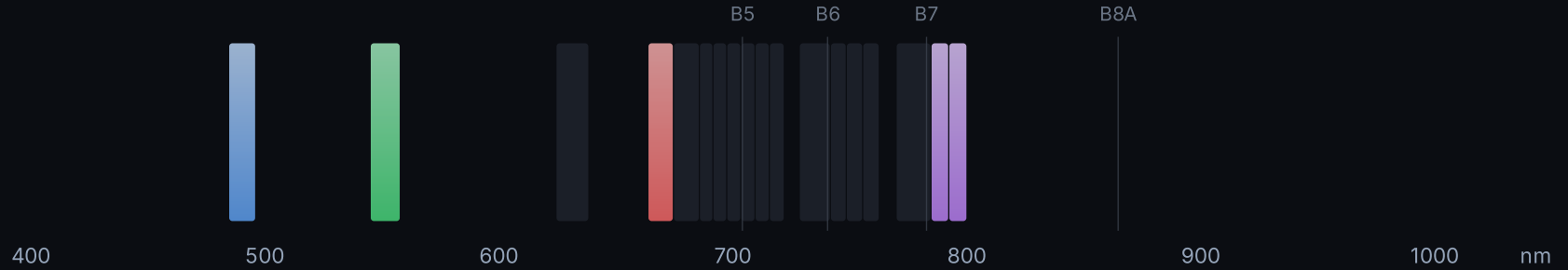


B8
780-886 nm

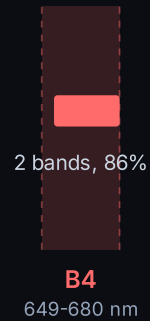
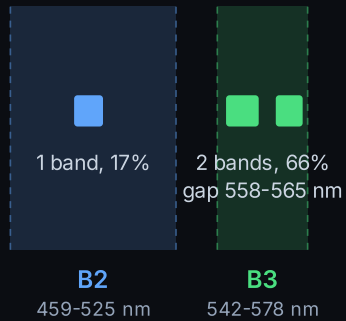
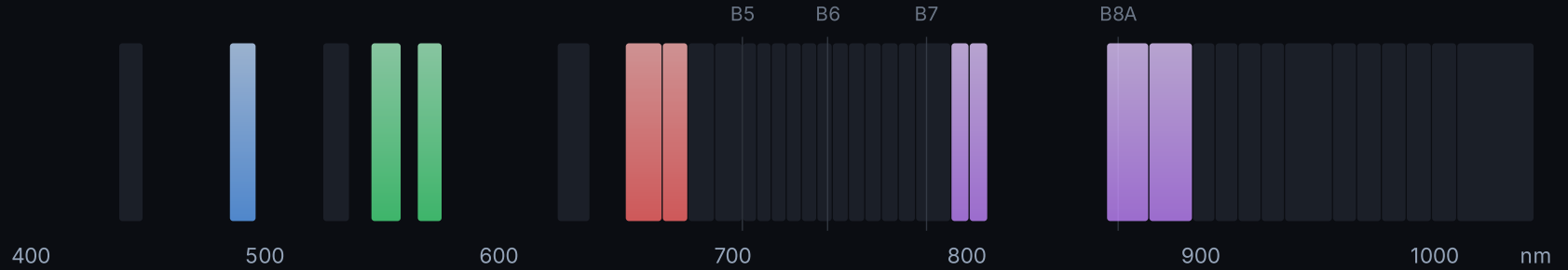
Mode 3 : 18 bands, ~13 m



Mode 4 : 18 bands, ~11 m



Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Where the bands end up

R, B4

G, B3

B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	42-51 (10), 782-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	16-17 (2), 771-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	14-15 (2), 773-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	17-18 (2), 785-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1 (1), 469-500	2 (1), 500-593	3 (1), 609-689	4 (1), 782-901

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.